

Practical NO. 4

Aggregate Functions with GROUP BY and HAVING

```
EMPLOYEE (  
    Emp_ID INT PRIMARY KEY,  
    Emp_Name VARCHAR(50),  
    Department VARCHAR(50),  
    Salary INT,  
    City VARCHAR(50)  
);
```

```
mysql> SELECT * FROM EMPLOYEE;  
+-----+-----+-----+-----+-----+  
| Emp_ID | Emp_Name | Department | Salary | City |  
+-----+-----+-----+-----+-----+  
|      1 | Amit    | IT         | 50000 | Pune |  
|      2 | Neha    | HR         | 60000 | Mumbai |  
|      3 | Rahul   | IT         | 40000 | Pune |  
|      4 | Sneha   | Finance    | 50000 | Mumbai |  
|      5 | Ankit   | HR         | 45000 | Delhi |  
+-----+-----+-----+-----+-----+  
5 rows in set (0.00 sec)
```

Basic Aggregate Functions

Count total employees

```
SELECT COUNT(*) AS Total_Employees  
FROM EMPLOYEE;
```

```
mysql> SELECT COUNT(*) AS Total_Employee  
-> FROM EMPLOYEE;  
+-----+  
| Total_Employee |  
+-----+  
|          5 |  
+-----+  
1 row in set (0.00 sec)
```

Find average salary

```
SELECT AVG(Salary) AS Avg_Salary  
FROM EMPLOYEE;
```

```
mysql> SELECT AVG(Salary) AS Avg_Salary
-> FROM EMPLOYEE;
+-----+
| Avg_Salary |
+-----+
| 49000.0000 |
+-----+
1 row in set (0.00 sec)
```

Find maximum salary

```
SELECT MAX(Salary) AS Highest_Salary
FROM EMPLOYEE;
```

```
mysql> SELECT MAX(Salary) AS Highest_Salary
-> FROM EMPLOYEE;
+-----+
| Highest_Salary |
+-----+
|          60000 |
+-----+
1 row in set (0.00 sec)
```

Find minimum salary

```
SELECT MIN(Salary) AS Lowest_Salary
FROM EMPLOYEE;
```

```
mysql> SELECT MIN(Salary) As Lowest_Salary
-> FROM EMPLOYEE;
+-----+
| Lowest_Salary |
+-----+
|          40000 |
+-----+
1 row in set (0.00 sec)
```

Find total salary paid

```
SELECT SUM(Salary) AS Total_Salary
FROM EMPLOYEE;
```

```
mysql> SELECT SUM(Salary) AS Total_Salary
-> FROM EMPLOYEE;
+-----+
| Total_Salary |
+-----+
|      245000 |
+-----+
1 row in set (0.00 sec)
```

Using GROUP BY Clause

Count employees department-wise

```
SELECT Department, COUNT(*) AS Total_Employees
FROM EMPLOYEE
GROUP BY Department;
```

```
mysql> SELECT Department, COUNT(*) AS Total_Employees
-> FROM EMPLOYEE
-> GROUP BY Department;
+-----+-----+
| Department | Total_Employees |
+-----+-----+
| Finance    |                1 |
| HR         |                2 |
| IT         |                2 |
+-----+-----+
3 rows in set (0.00 sec)
```

Find average salary department-wise

```
SELECT Department, AVG(Salary) AS Avg_Salary
FROM EMPLOYEE
GROUP BY Department;
```

```
mysql> SELECT Department, AVG(Salary) AS Avg_Salary
-> FROM EMPLOYEE
-> GROUP BY Department;
+-----+-----+
| Department | Avg_Salary |
+-----+-----+
| Finance    | 50000.0000 |
| HR         | 52500.0000 |
| IT         | 45000.0000 |
+-----+-----+
3 rows in set (0.00 sec)
```

Find total salary department-wise

```
SELECT Department, SUM(Salary) AS Total_Salary
FROM EMPLOYEE
GROUP BY Department;
```

```
mysql> SELECT Department, SUM(Salary) AS Total_Salary
-> FROM EMPLOYEE
-> GROUP BY Department;
+-----+-----+
| Department | Total_Salary |
+-----+-----+
| Finance    |          50000 |
| HR         |         105000 |
| IT         |          90000 |
+-----+-----+
3 rows in set (0.00 sec)
```

Find maximum salary city-wise

```
SELECT City, MAX(Salary) AS Highest_Salary
FROM EMPLOYEE
GROUP BY City;
```

```
mysql> SELECT City, MAX(Salary) AS Highest_Salary
-> FROM EMPLOYEE
-> GROUP BY City;
+-----+-----+
| City    | Highest_Salary |
+-----+-----+
| Delhi   |          45000 |
| Mumbai  |         60000 |
| Pune    |          50000 |
+-----+-----+
3 rows in set (0.00 sec)
```

Using HAVING Clause

Departments having more than 5 employees

```
SELECT Department, COUNT(*) AS Total_Employees
FROM EMPLOYEE
GROUP BY Department
HAVING COUNT(*) > 5;
```

```
mysql> SELECT Department, COUNT(*) AS Total_Employees
-> FROM EMPLOYEE
-> GROUP BY Department
-> HAVING COUNT(*) > 5;
Empty set (0.00 sec)
```

Departments where average salary is greater than 50000

```
SELECT Department, AVG(Salary) AS Avg_Salary
FROM EMPLOYEE
GROUP BY Department
HAVING AVG(Salary) > 50000;
```

```
mysql> SELECT Department, COUNT(*) AS Avg_Salary
-> FROM EMPLOYEE
-> GROUP BY Department
-> HAVING AVG(Salary) > 50000;
+-----+-----+
| Department | Avg_Salary |
+-----+-----+
| HR          |          2 |
+-----+-----+
1 row in set (0.00 sec)
```

Cities where total salary is greater than 40,000

```
SELECT City, SUM(Salary) AS Total_Salary
FROM EMPLOYEE
GROUP BY City
HAVING SUM(Salary) > 40000;
```

```
mysql> SELECT City, SUM(Salary) AS Total_Salary
-> FROM EMPLOYEE
-> GROUP BY City
-> HAVING SUM(Salary) > 40000;
+-----+-----+
| City    | Total_Salary |
+-----+-----+
| Delhi   |          45000 |
| Mumbai  |         110000 |
| Pune    |          90000 |
+-----+-----+
3 rows in set (0.00 sec)
```

GROUP BY with WHERE and HAVING Together

Average salary of employees from Pune department-wise

```
SELECT Department, AVG(Salary) AS Avg_Salary
FROM EMPLOYEE
WHERE City = 'Pune'
GROUP BY Department
HAVING AVG(Salary) > 40000;
```

```
mysql> SELECT Department, AVG(Salary) AS Avg_Salary
-> FROM EMPLOYEE
-> WHERE City = 'Pune'
-> GROUP BY Department
-> HAVING AVG(Salary) > 40000;
+-----+-----+
| Department | Avg_Salary |
+-----+-----+
| IT         | 45000.0000 |
+-----+-----+
1 row in set (0.00 sec)
```

Find department with highest average salary

```
SELECT Department, AVG(Salary) AS Avg_Salary
FROM EMPLOYEE
GROUP BY Department
ORDER BY Avg_Salary DESC
LIMIT 1;
```

```
mysql> SELECT Department, AVG(Salary) AS Avg_Salary
-> FROM EMPLOYEE
-> GROUP BY Department
-> ORDER BY Avg_Salary DESC
-> LIMIT 1;
+-----+-----+
| Department | Avg_Salary |
+-----+-----+
| HR         | 52500.0000 |
+-----+-----+
1 row in set (0.01 sec)
```